

Lenses and Images

Young's Double-Slit Experiment

Converging and diverging lenses, the lens equation and magnification, and the interference of light through two narrow slits.

Grade 11 · Egyptian Baccalaureate

31 questions

Step-by-step solutions

 $\Delta y = \lambda L / d$

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PHYSICS

The toolkit for these two lessons

$$1/f = 1/a + 1/b$$

the lens equation, a = object distance

$$M = b/a = h'/h$$

the linear magnification

$$b > 0 \rightarrow \text{real, inverted}$$

$b < 0 \rightarrow$ virtual, erect

$$f > 0 \text{ convex} \cdot f < 0 \text{ concave}$$

the sign of the focal length

$$\Delta y = \lambda L / d$$

the fringe separation in Young's experiment

$$\Delta = m\lambda \rightarrow \text{bright}$$

$\Delta = (m + \frac{1}{2})\lambda \rightarrow \text{dark}$

Two lessons, two pictures. A lens bends **one** beam so that it meets again at an image — everything there comes out of $1/f = 1/a + 1/b$. Young's experiment splits **one** beam into two and lets them meet again as fringes — everything there comes out of the path difference. Keep the two pictures apart and the whole unit becomes easy.

CLASSWORK 1

A Lesson 2–10 · the parts of a lens

1 What is the name of the lens that is thicker at its centre than at its edges?

EXPLAIN

2 The point at which rays parallel to the principal axis collect after being refracted by a convex lens is called:

MCQ

A the optical centre

B the principal focus

C the centre of curvature

D the pole of the lens

3 The straight line that passes through the optical centre of the lens and is perpendicular to its plane is known as:

MCQ

A the focal length

B the principal axis

C the radius

D the incident ray

Lesson 2–10 · images formed by lenses

- 4 Using the ray diagram opposite, explain why the image formed by a concave lens is always virtual, erect and diminished.

EXPLAIN

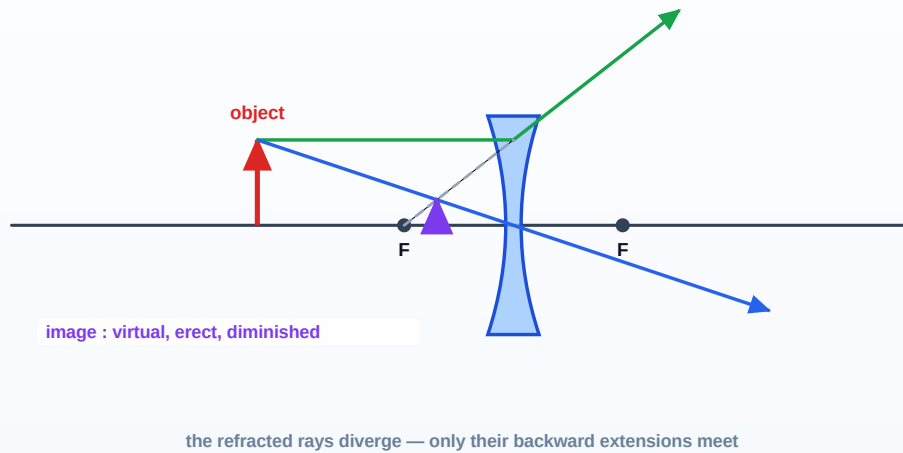


FIGURE 1 · A DIVERGING (CONCAVE) LENS

- 5 Draw the path of the rays for a magnifying (convex) lens used to look at an object placed at a small distance from it (inside the focal length), showing the ray that passes through the optical centre, the ray parallel to the principal axis, and how their extensions meet.

EXPLAIN

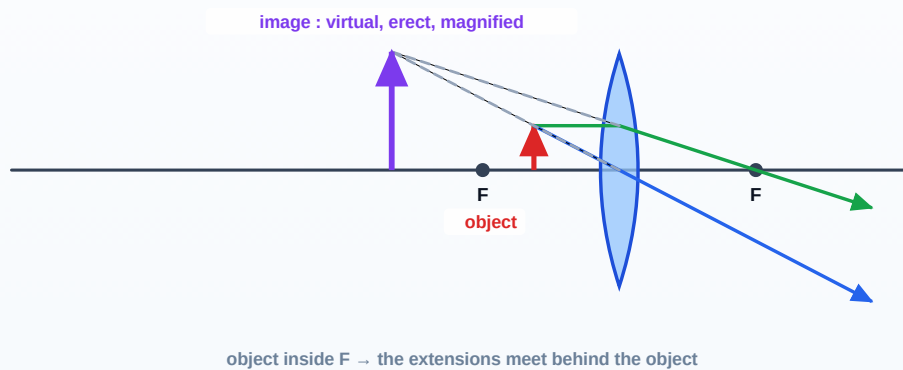


FIGURE 2 · A CONVEX LENS USED AS A MAGNIFIER

- 6 An object is placed at a distance a from a convex lens and a virtual image is formed. The lens is then replaced by a concave lens having the **same focal length (f)**, without moving the object, and another virtual image is formed. If the distance between the position of the first image and the position of the second image equals the focal length f , **find the object distance a in terms of f .**

PROBLEM

- 7 If an object is placed at a distance much greater than twice the focal length ($2F$) of a convex lens, the image formed for it is:

MCQ

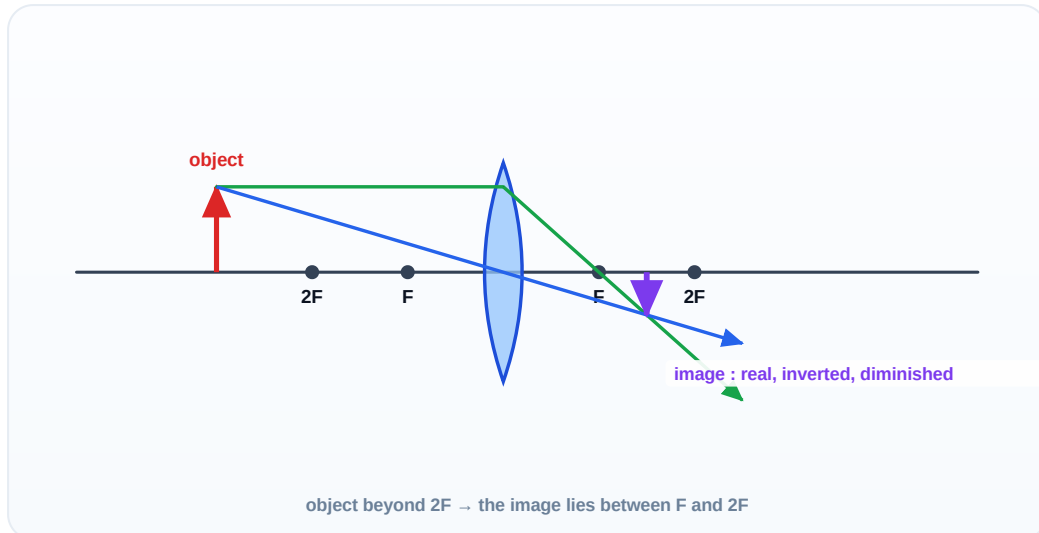


FIGURE 3 · AN OBJECT BEYOND $2F$

- ☐ A real, inverted, diminished
- ☐ B real, inverted, magnified
- ☐ C virtual, erect, diminished
- ☐ D real, inverted, equal to the object

- 8 If the magnification ($M = 0.5$), this means that the image is:

MCQ

- ☐ A magnified to twice the object
- ☐ B equal to the object
- ☐ C diminished to half the object
- ☐ D always inverted

Lesson 2–11 · Young's experiment

- 9 The figure opposite shows Young's double-slit experiment. If the distance between two successive bright fringes is **5 mm**, **calculate the wavelength of the light used.**

PROBLEM

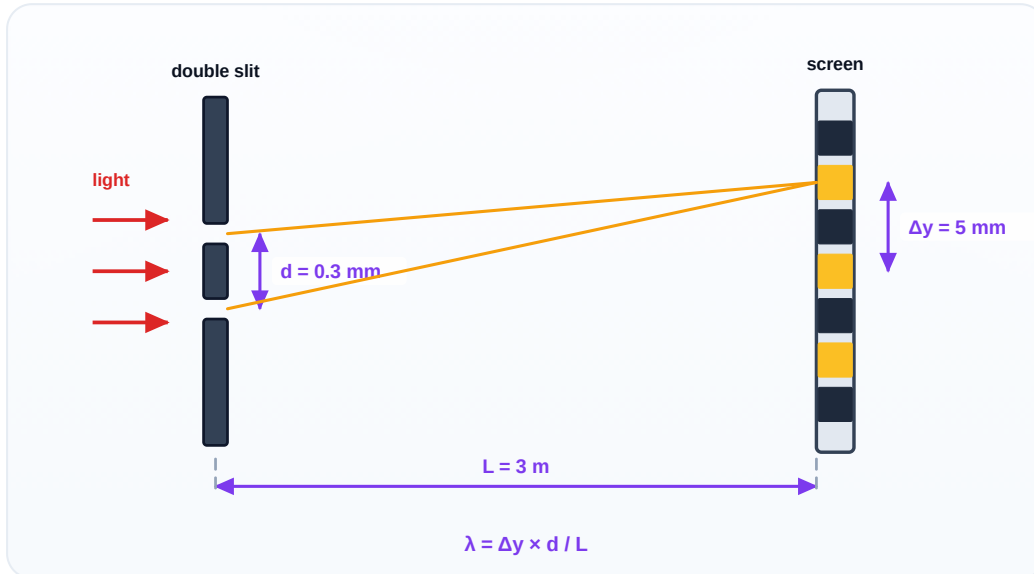


FIGURE 4 · THE DOUBLE-SLIT ARRANGEMENT

- 10 The wavelength λ of any monochromatic light in Thomas Young's double-slit experiment is determined from the relation (R is the distance from the double slit to the screen)

MCQ

A $\lambda = \Delta y R / d$

B $R = \Delta y d / \lambda$

C $\Delta y = \lambda d / R$

D $d = \lambda \Delta y / R$

- 11 A monochromatic ray falls in Thomas Young's experiment and the distance between the two openings of the double slit is d_1 . The double slit is then replaced by another one whose slit separation is **half** the first. The distance between two successive fringes of the same kind in the second case is:

MCQ

A $\Delta y_1 / 2 = \Delta y_2$

B $\Delta y_1 = \Delta y_2$

C $4 \Delta y_1 = \Delta y_2$

D $2 \Delta y_1 = \Delta y_2$

Lesson 2–11 · fringes and their separation

- 12 Monochromatic light of wavelength 450 nm passes through two parallel slits separated by 0.3 mm, and the bright fringes can be observed on a screen 2 m away. **Calculate, in millimetres, the distance between two successive fringes of the same kind.**

PROBLEM

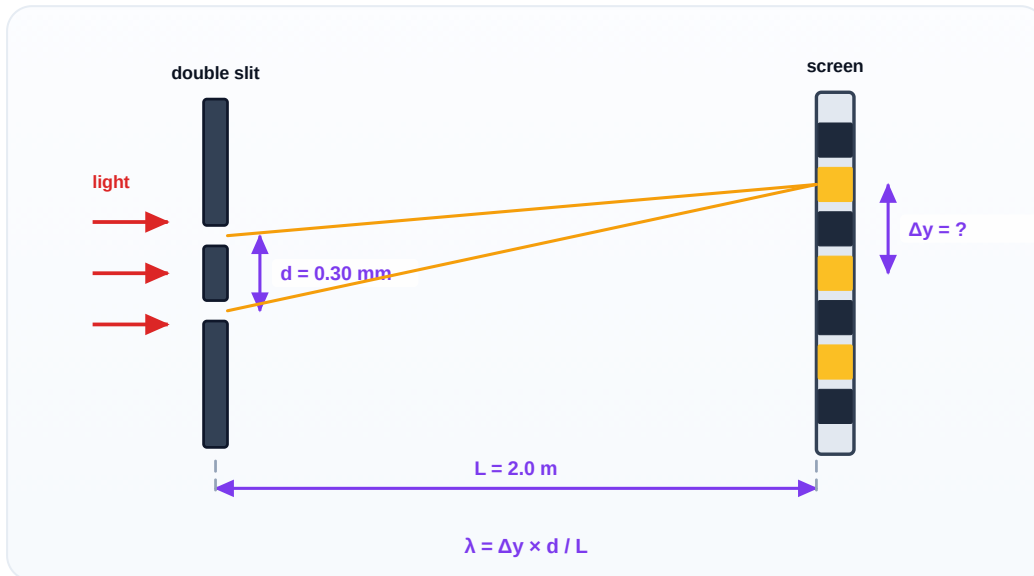


FIGURE 5 · NOT TO SCALE

- 13 From the figure opposite, which represents Young's double-slit experiment: with both slits open, **will two bright patches appear on the wall — one for each slit — or a row of many bright fringes? And where is the brightest one?**

EXPLAIN

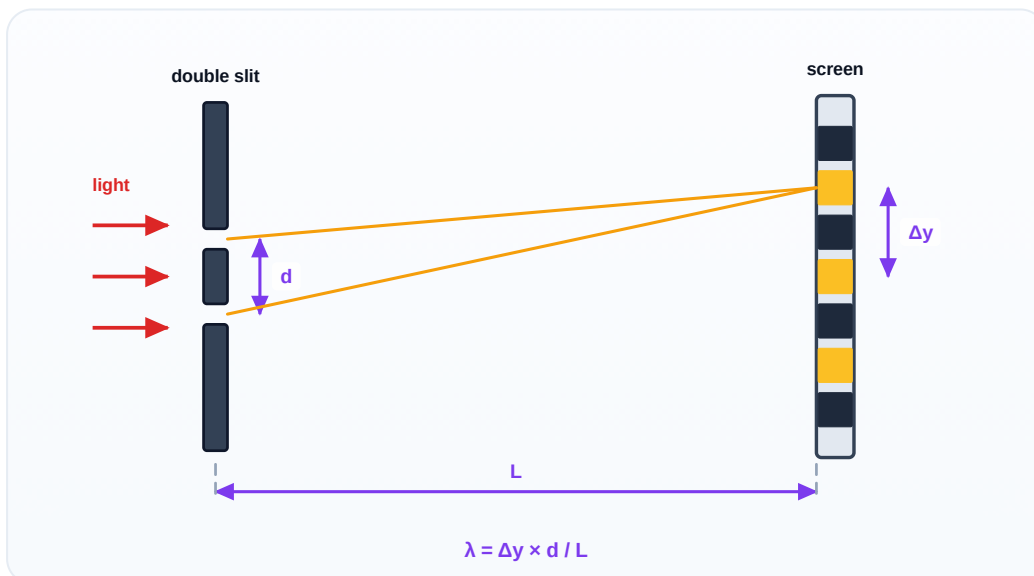


FIGURE 6 · BOTH SLITS OPEN

- 14 Monochromatic light of wavelength 600 nm was used in Young's experiment, and interference fringes were formed on a screen 1.2 m away. If the separation between successive bright fringes is 2.4 mm, **calculate the distance (d) between the two slits.**

PROBLEM

15 Which of the following methods can increase the distance between the interference fringes in Young's double-slit experiment, using a green monochromatic light source?

MCQ

A decreasing the distance between the screen that receives the fringes and the two narrow slits

B increasing the distance between the source and the two narrow slits

C increasing the separating distance between the two narrow slits

D replacing the light source with a red monochromatic light source

16 In Thomas Young's experiment, when the distance between the double-slit screen and the screen prepared to receive the fringes is doubled, then the distance between every two successive fringes of the same kind

MCQ

A doubles and the clarity of the fringes decreases

B doubles and the clarity of the fringes increases

C halves and the clarity of the fringes increases

D halves and the clarity of the fringes decreases

- 17 A convex lens whose focal length is 10 cm; an object is placed at a distance of 15 cm from it. The image formed is:

MCQ

- ☐ A virtual, magnified
- ☐ B real, magnified
- ☐ C real, diminished
- ☐ D real, equal to the object

- 18 Four students were allowed to choose among several sources of different wavelength, several double slits, and a suitable screen distance, in order to carry out the double-slit experiment. For which student will the fringes be the clearest (the most widely separated)?

MCQ

	the wavelength	the distance between the two coherent sources	the distance of the screen from the double slit
A	500 nm	2 d	0.5 m
B	400 nm	d	1 m
C	600 nm	3 d	1.5 m
D	450 nm	0.5 d	1.1 m

- ☐ A choice A
- ☐ B choice B
- ☐ C choice C
- ☐ D choice D

- 19 A student carries out an experiment to demonstrate the interference of light using the double-slit experiment as in the figure, but he found that the interference fringes observed on the screen are so close together that they cannot be told apart. **What changes could help the student to distinguish between the fringes? (two are enough)**

EXPLAIN

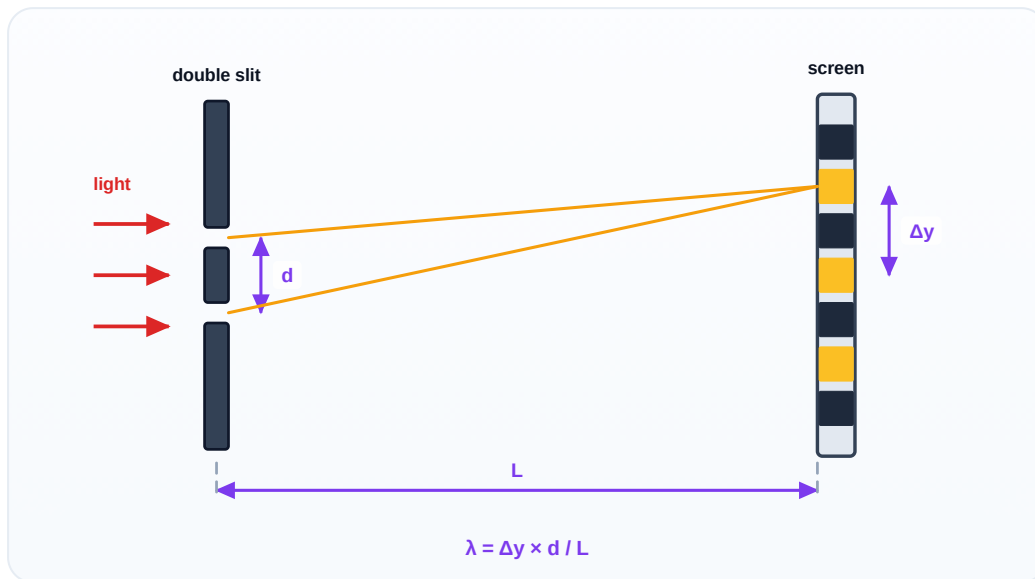


FIGURE 7 · THE FRINGES ARE TOO CLOSE TOGETHER

- 20 An object of length h is placed between a screen and a convex lens. When the position of the lens is changed, it is found that there are two positions that give a clear image on the screen. If the magnification at the first position is $M_1 = 4$ and the magnification at the second position is $M_2 = 0.25$, **calculate the length of the first image and of the second image.**

PROBLEM

- 21 **How can you increase the distance between the interference fringes** in Young's double-slit experiment, using a green monochromatic light source?

EXPLAIN

REVIEW

F

Group B

- 22 An image is formed on a wall by a lens. If the distance between the lens and the wall is 30 cm and the magnification is $M = 2$, then **the type of the lens and the object distance** are:

MCQ

A concave, $a = 15$ cm

B convex, $a = 60$ cm

C convex, $a = 15$ cm

D concave, $a = 60$ cm

- 23 In an experiment on the interference of light by Young, the distance between the two narrow slits is 0.2 mm, the wavelength of the monochromatic light used is 600 nm, and the distance between the two narrow slits and the screen prepared to receive the fringes is 1 m. Then the distance between two successive fringes of the same kind equals

MCQ

A 3 mm

B 6 mm

C 30 mm

D 1 m

- 24 A convex lens gives, for an object placed in front of it, an erect image magnified 4 times. The object is then moved away from the lens through a distance of 6 cm, and the image becomes real, inverted and magnified 4 times as well. **Calculate the focal length of the lens.**

PROBLEM

- 25 A screen is placed at a distance of 90 cm from an object, and a convex lens of focal length 20 cm moves between the object and the screen. **Find all the values of the distance a between the lens and the object at which a clear image is formed on the screen.**

PROBLEM

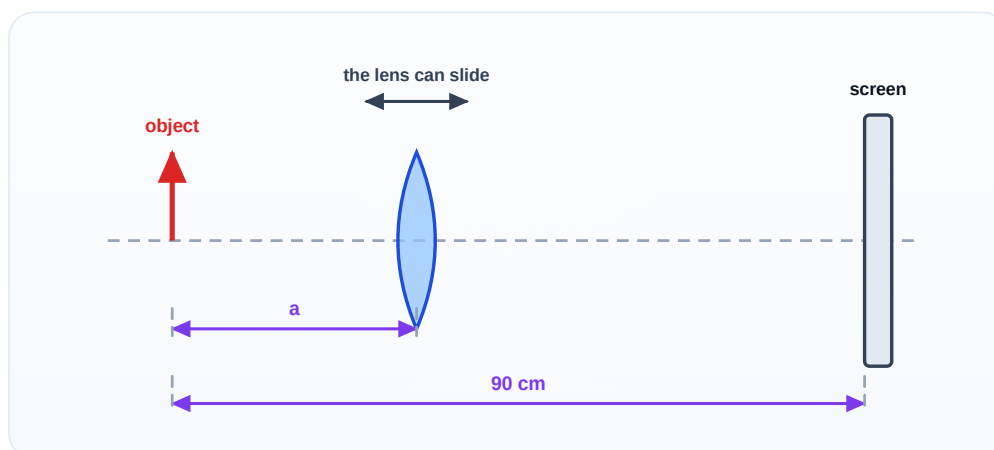


FIGURE 8 · THE LENS SLIDES BETWEEN OBJECT AND SCREEN

- 26 The drawing opposite shows the paths of two waves from the two slits S_1 and S_2 meeting at a point P. If the wavelength is λ and the distance S_2P exceeds S_1P by 1.5λ , **what type of fringe is formed at the point P? And why?**

EXPLAIN

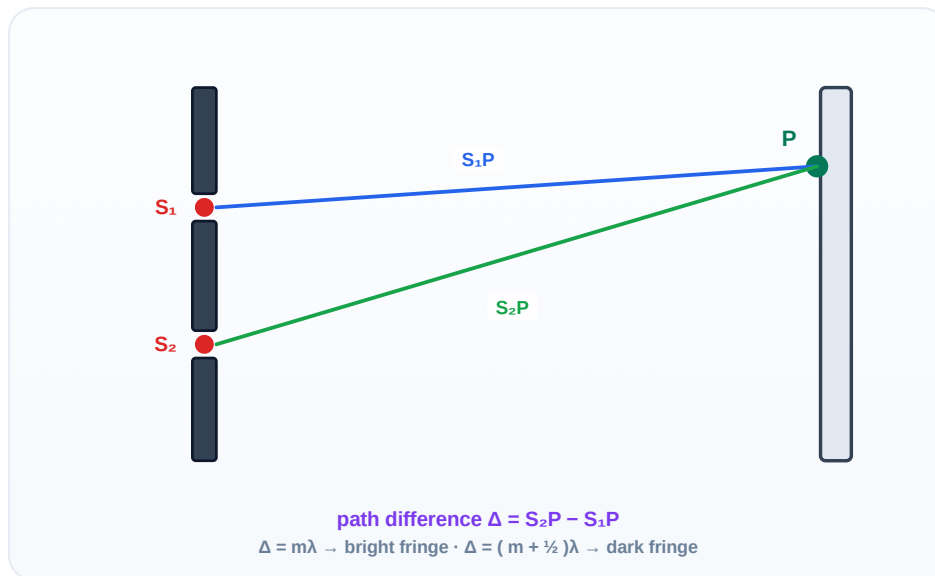


FIGURE 9 · THE TWO PATHS TO THE POINT P

REVIEW

G

Group C

- 27 When red light is used instead of blue light in Young's experiment, the distance between the fringes:

MCQ

- ☐ A increases, because the wavelength of red is larger
- ☐ B decreases, because the wavelength of red is smaller
- ☐ C does not change
- ☐ D the fringes disappear

- 28 A concave lens whose focal length is 20 cm; an object is placed at a distance of 20 cm from it. The image is:

MCQ

- ☐ A at infinity
- ☐ B virtual, erect, diminished
- ☐ C real, equal to the object
- ☐ D virtual, magnified

- 29 In the double-slit experiment, monochromatic light of wavelength $6000 \text{ \AA}$ was used and fringes were formed on a screen at a distance (R) from the double slit, the distance between every two successive bright fringes being Δy_1 . If monochromatic light of wavelength $4000 \text{ \AA}$ is then used and the distance between the double slit and the screen is doubled, the distance between every two successive bright fringes becoming Δy_2 , **find the ratio $\Delta y_1 / \Delta y_2$.**

PROBLEM

30 An object is placed beyond $2F$ of a convex lens, then it moves with a uniform velocity v towards the focus F . **Prove that the speed of the image formed is variable, and that it increases as the object approaches the focus.**

PROBLEM

31 **Give the scientific reason:** the central fringe in Young's double-slit experiment is always a bright fringe.

EXPLAIN



Solutions — step by step

Two equations run this whole handout. For every lens question write $1/f = 1/a + 1/b$ and $M = b/a$, put f negative for a concave lens and b negative for a virtual image, and the algebra does the rest. For every Young question write $\Delta y = \lambda L/d$ and ask only which of the three letters on the right has changed.

Part A — Classwork 1

1 the lens that is thicker at the centre

a lens that is thicker at its centre than at its edges is the
convex (converging) lens

parallel rays that strike it are bent inwards and really meet at a point,
so its principal focus is a real focus and its focal length is positive.

The mirror image of this rule is just as useful: a lens that is **thinner** at its centre than at its edges is the concave (diverging) lens, its rays spread apart, its focus is virtual and its focal length is taken as negative.

the convex (converging) lens

2 where parallel rays collect

rays parallel to the principal axis are refracted by a convex lens and meet
at one point on the axis ; that point is the principal focus (F).
the distance from the optical centre to F is the focal length f.

Do not confuse the three points on the axis. The **optical centre** is the middle of the lens itself and a ray through it goes straight on; the **centre of curvature** is the centre of the sphere the surface was cut from; the **principal focus** is where the parallel rays actually gather.

choice B — the principal focus

3 the line through the optical centre

the straight line through the optical centre, perpendicular to the plane of
the lens, is the principal axis.

it is the reference line from which every distance (a , b , f , 2f) is measured.

Note on the printed lesson. In the book the questions of this classwork are numbered 1, 3, 4 — number 2 is missing from the printed page. Nothing is lost; the numbering here has simply been made continuous.

choice B — the principal axis

Part B — Home assignment 1

4 why a concave lens always gives a virtual, erect, diminished image

take the two standard rays from the top of the object :

ray 1 : parallel to the axis, then refracted so that it seems to come from the principal focus F on the same side as the object.

ray 2 : straight through the optical centre, undeviated.

after the lens the two rays diverge , so they can never really cross :

→ only their backward extensions meet → the image is VIRTUAL

both extensions meet above the axis, on the same side as the object :

→ the image is ERECT

algebraically, with f negative and a positive :

$$1/b = -1/f - 1/a \quad \rightarrow \quad b = -af / (a + f)$$

$$M = |b| / a = f / (a + f) < 1 \quad \text{for every positive } a$$

→ the image is DIMINISHED , and it always lies between F and the lens

The last line is the real proof: $f/(a+f)$ is a fraction whose denominator is bigger than its numerator for **any** positive a, so M can never reach 1. This is exactly why a diverging lens is used in spectacles for short sight, and in a door viewer — it shrinks a wide scene into a small erect picture.

always virtual, erect and diminished — whatever the object distance

5 the ray diagram of the magnifying glass

the object is placed inside the focal length : $a < f$

ray 1 : parallel to the axis, then refracted through the far focus F'.

ray 2 : straight through the optical centre O.

after the lens the two refracted rays are still diverging , so they are traced backwards (dashed) until their extensions meet.

$$1/b = 1/f - 1/a, \quad \text{and} \quad a < f \quad \rightarrow \quad b \text{ is negative}$$

$$b = -af / (f - a) \quad M = f / (f - a) > 1$$

The closer the object creeps towards F, the smaller $(f - a)$ becomes and the larger M grows — which is exactly what you feel when you slide a magnifying glass until the print suddenly leaps in size just before it blurs.

position of the object	position of the image	nature of the image
beyond 2F	between F and 2F	real, inverted, diminished
at 2F	at 2F	real, inverted, equal
between F and 2F	beyond 2F	real, inverted, magnified
at F	at infinity	no image is formed
inside F	same side as the object	virtual, erect, magnified

the extensions meet on the object side → a virtual, erect, magnified image

6 the two virtual images a focal length apart

convex lens, virtual image (so $a < f$) :

$$1/b_1 = 1/f - 1/a = (a - f) / (a f)$$

$$|b_1| = a f / (f - a) \quad \text{on the object side}$$

concave lens of the same f (f is negative) :

$$1/b_2 = -1/f - 1/a = -(a + f) / (a f)$$

$$|b_2| = a f / (a + f) \quad \text{also on the object side}$$

both images lie on the same side, so the distance between them is

$$\begin{aligned} |b_1| - |b_2| &= a f / (f - a) - a f / (a + f) \\ &= a f [(a + f) - (f - a)] / (f - a)(f + a) \\ &= 2 a^2 f / (f^2 - a^2) \end{aligned}$$

this distance is given equal to f :

$$2 a^2 f / (f^2 - a^2) = f$$

$$2 a^2 = f^2 - a^2$$

$$3 a^2 = f^2$$

$$a = f / \sqrt{3} = (\sqrt{3} / 3) f \approx 0.58 f$$

The answer had to come out smaller than f , and it does — a convex lens can only make a virtual image when the object is inside the focus. Notice also that the convex image is the farther of the two: $a f / (f - a)$ is always bigger than $a f / (a + f)$, which is why the subtraction was taken in that order.

$$a = f / \sqrt{3} \approx 0.58 f$$

7 an object far beyond $2F$

$$1/b = 1/f - 1/a, \quad \text{with } a \text{ very large}$$

$$1/a \rightarrow \text{very small} \rightarrow b \rightarrow f$$

so the image settles just outside the focus, between F and $2F$:

$$M = b / a \approx f / a \ll 1$$

b is positive $\rightarrow$ the image is real and inverted

This is the camera and the eye. A very distant object throws its image practically **at** the focus, tiny and upside down — which is why the sensor of a camera sits one focal length behind the lens when you photograph a far-away mountain.

position of the object	position of the image	nature of the image
beyond $2F$	between F and $2F$	real, inverted, diminished
at $2F$	at $2F$	real, inverted, equal
between F and $2F$	beyond $2F$	real, inverted, magnified
at F	at infinity	no image is formed
inside F	same side as the object	virtual, erect, magnified

choice A — real, inverted and diminished

8 the meaning of $M = 0.5$

$$M = h' / h = 0.5 \quad \rightarrow \quad h' = 0.5 h$$

the image height is half the object height.

M answers only one question: **how big**. Whether the image is erect or inverted is carried by the **sign**, and the printed $M = 0.5$ carries no sign, so nothing at all can be said here about the orientation — which is why choice D is wrong.

choice C — the image is diminished to half the object

Part C — Classwork 2

9 the wavelength from the fringe separation

from the figure : $d = 0.3 \text{ mm} = 0.3 \times 10^{-3} \text{ m}$, $L = 3 \text{ m}$

given : $\Delta y = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$

$$\Delta y = \lambda L / d \quad \rightarrow \quad \lambda = \Delta y \cdot d / L$$

$$\lambda = (5 \times 10^{-3} \times 0.3 \times 10^{-3}) / 3$$

$$\lambda = (1.5 \times 10^{-6}) / 3$$

$$\lambda = 5 \times 10^{-7} \text{ m} = 500 \text{ nm}$$

A sanity check you should always make: visible light runs from about 400 nm (violet) to about 700 nm (red). An answer of 500 nm sits comfortably in the middle, in the green. If a double-slit calculation ever hands you microns or nanometres by the thousand, a power of ten has slipped.

$\lambda = 5 \times 10^{-7} \text{ m} = 500 \text{ nm}$ (green light)

10 which relation gives λ

the fringe separation in Young's experiment is

$$\Delta y = \lambda R / d$$

rearranged, the three correct forms are

$$\lambda = \Delta y \cdot d / R$$

$$R = \Delta y \cdot d / \lambda$$

$$d = \lambda R / \Delta y$$

testing the four printed choices :

A) $\lambda = \Delta y R / d$ $\times$ (R and d are interchanged)

B) $R = \Delta y d / \lambda$ $\checkmark$

C) $\Delta y = \lambda d / R$ $\times$ (d and R are interchanged)

D) $d = \lambda \Delta y / R$ $\times$ (Δy and R are interchanged)

Note on the printed question. The wording asks for the relation that gives λ , which points at choice A — but A has R and d the wrong way round and is wrong. The only sound statement among the four is B. Keep the master form $\Delta y = \lambda R / d$ in your head and read the others off it: λ and R are always **upstairs** together, d is always **downstairs**.

choice B — it is the only algebraically correct relation of the four

11 halving the slit separation

$$\Delta y = \lambda R / d \quad \rightarrow \quad \Delta y \propto 1 / d \quad (\lambda \text{ and } R \text{ unchanged})$$

$$d_2 = d_1 / 2$$

$$\Delta y_2 / \Delta y_1 = d_1 / d_2 = d_1 / (d_1 / 2) = 2$$

$$\Delta y_2 = 2 \Delta y_1$$

Inverse proportion is the trap here. Squeezing the two slits **closer** pushes the fringes **farther** apart — the same reason a very narrow gap spreads a wave over a wide angle. Everything that makes the geometry tighter makes the pattern wider.

choice D — $2 \Delta y_1 = \Delta y_2$, the fringes spread to twice their separation

Part D — Home assignment 2

12 the fringe separation for 450 nm

$$\lambda = 450 \text{ nm} = 450 \times 10^{-9} \text{ m} = 4.5 \times 10^{-7} \text{ m}$$

$$d = 0.3 \text{ mm} = 3 \times 10^{-4} \text{ m}, \quad L = 2.0 \text{ m}$$

$$\Delta y = \lambda L / d$$

$$\Delta y = (4.5 \times 10^{-7} \times 2.0) / (3 \times 10^{-4})$$

$$\Delta y = (9 \times 10^{-7}) / (3 \times 10^{-4})$$

$$\Delta y = 3 \times 10^{-3} \text{ m} = 3 \text{ mm}$$

Three millimetres between fringes on a screen two metres away is typical for a school set-up, and it is why the slits have to be a fraction of a millimetre apart. Open them to a millimetre and Δy drops to under a millimetre — the pattern becomes a blur.

$$\Delta y = 3 \text{ mm}$$

13 two patches, or a row of fringes?

each slit is narrow enough to diffract the light, so the two beams spread out and overlap on the wall; being cut from one wavefront they are coherent (a constant phase difference).

wherever the path difference is $\Delta = m \lambda \rightarrow$ constructive $\rightarrow$ bright

wherever the path difference is $\Delta = (m + \frac{1}{2}) \lambda \rightarrow$ destructive $\rightarrow$ dark

these conditions repeat right across the wall, so the result is a row of many bright and dark fringes, not two separate patches.

at the centre (point 0) $S_1O = S_2O \rightarrow \Delta = 0 \rightarrow$ always bright

Two patches is what you would get from two **independent** lamps, because their phase difference would change millions of times a second and the fringes would wash out. The whole point of Young's single source lighting both slits is to make the two beams coherent.

a row of many bright fringes separated by dark ones, and the brightest is the central fringe opposite the midpoint between the two slits

14 finding the slit separation

$$\lambda = 600 \text{ nm} = 6 \times 10^{-7} \text{ m}, \quad L = 1.2 \text{ m}, \quad \Delta y = 2.4 \text{ mm} = 2.4 \times 10^{-3} \text{ m}$$

$$\Delta y = \lambda L / d \rightarrow d = \lambda L / \Delta y$$

$$d = (6 \times 10^{-7} \times 1.2) / (2.4 \times 10^{-3})$$

$$d = (7.2 \times 10^{-7}) / (2.4 \times 10^{-3})$$

$$d = 3 \times 10^{-4} \text{ m} = 0.3 \text{ mm}$$

Notice how small d has to be. The slit separation comes out about 500 wavelengths — and that is exactly why Young's experiment had to wait for careful ruling: any wider and the fringes crowd into an unreadable band.

$$d = 3 \times 10^{-4} \text{ m} = 0.3 \text{ mm}$$

15 how to widen the fringes

$$\Delta y = \lambda L / d$$

to make Δy bigger : increase λ , increase L , or decrease d .

checking the printed choices :

- A) decrease L → Δy decreases ✗
- B) source-to-slit distance does not appear in the law ✗
- C) increase d → Δy decreases ✗
- D) green → red , λ larger → Δy increases ✓

Red is the long-wavelength end of the visible spectrum (≈ 700 nm against ≈ 530 nm for green), so simply swapping the colour stretches the pattern by about a third. Choice B is the distractor worth naming: the distance from the lamp to the slits changes the **brightness**, never the fringe spacing.

choice D — replace the green source by a red one

16 doubling the screen distance

$$\Delta y = \lambda L / d \quad \rightarrow \quad \Delta y \propto L$$

$$L_2 = 2 L_1 \quad \rightarrow \quad \Delta y_2 = 2 \Delta y_1$$

but the same amount of light is now spread over twice the area, so each fringe is fainter and its edges are less sharp :

→ the separation doubles , the clarity (contrast) decreases

This is the trade-off every experimenter meets: you can always spread a pattern out, but you pay for the space with brightness. Doubling L spreads the same energy over four times the area, so the fringes are about four times dimmer.

choice A — the separation doubles and the clarity of the fringes decreases

Part E — Review, Group A

17 convex lens, $f = 10$ cm, $a = 15$ cm

$$1/f = 1/a + 1/b$$

$$1/b = 1/10 - 1/15 = (3 - 2) / 30 = 1 / 30$$

$$b = + 30 \text{ cm} \quad (\text{ positive } \rightarrow \text{ real, on the far side })$$

$$M = b / a = 30 / 15 = 2 \quad (\text{ magnified })$$

Read the position first and the answer follows without any arithmetic: $f = 10$ so $2F = 20$, and the object at 15 cm sits **between F and 2F** — the one region that always gives a real, inverted, magnified image. This is the projector setting.

position of the object	position of the image	nature of the image
beyond 2F	between F and 2F	real, inverted, diminished
at 2F	at 2F	real, inverted, equal
between F and 2F	beyond 2F	real, inverted, magnified
at F	at infinity	no image is formed
inside F	same side as the object	virtual, erect, magnified

choice B — a real, magnified (and inverted) image, 30 cm from the lens

18 which student gets the clearest fringes

the fringes are easiest to tell apart when $\Delta y = \lambda L / d$ is largest.

taking d as the common unit :

A) $\Delta y = 500 \times 10^{-9} \times 0.5 / (2 d) = 1.25 \times 10^{-7} / d$

B) $\Delta y = 400 \times 10^{-9} \times 1.0 / (1 d) = 4.00 \times 10^{-7} / d$

C) $\Delta y = 600 \times 10^{-9} \times 1.5 / (3 d) = 3.00 \times 10^{-7} / d$

D) $\Delta y = 450 \times 10^{-9} \times 1.1 / (0.5 d) = 9.90 \times 10^{-7} / d \quad \leftarrow \text{the largest}$

Do not be tempted by the longest wavelength alone. Choice C has the reddest light of the four and still loses, because its slits are six times farther apart than D's. In $\lambda L/d$ all three letters vote, and here the small d carries the day.

choice D — 450 nm, 0.5 d, 1.1 m gives about 2.5 times the fringe separation of choice B

19 the fringes are too close together

$$\Delta y = \lambda L / d$$

so Δy can be made larger by

1. moving the screen farther from the double slit (increase L) ,
2. using a double slit whose two openings are closer (decrease d) ,
3. using light of a longer wavelength – swap the red filter for nothing less than red, i.e. use the reddest source available (increase λ).

The figure shows a red filter already in place, so the wavelength is close to the visible maximum and little more can be won there. The two changes with real room to move are L and d — which is why they are the two answers a marker will be looking for.

any two of: increase the slit–screen distance L , use a double slit with a smaller separation d , or use light of a longer wavelength λ

20 the two lens positions

the magnification is defined by

$$M = h' / h \quad \rightarrow \quad h' = M h$$

first position : $h'_1 = M_1 h = 4 h$

second position : $h'_2 = M_2 h = 0.25 h = h / 4$

The pair is no accident: for a fixed object–screen distance the two lens positions are mirror images of each other (object and image distances simply swap), so the two magnifications are always reciprocals — 4 and $\frac{1}{4}$. Multiply them and you get exactly 1, which is a quick check on any answer of this type.

the first image is 4 h long and the second is 0.25 h long

21 widening the fringes with a green source

$$\Delta y = \lambda L / d$$

to increase Δy , change any one of the three :

- increase L – move the screen farther from the double slit ,
- decrease d – use a double slit with the two openings closer together ,
- increase λ – replace the green source with a red one
($\lambda_{\text{red}} \approx 700 \text{ nm} > \lambda_{\text{green}} \approx 530 \text{ nm}$).

The one thing that will **not** work is turning the lamp up. Intensity changes how bright the fringes are, not where they fall; the geometry of $\Delta y = \lambda L/d$ contains no term for the power of the source.

increase L , decrease d , or move to a longer wavelength (green $\rightarrow$ red)

22 the image on the wall

an image caught on a wall is a real image, and only a convex lens can make one → the lens is convex.

$$b = 30 \text{ cm} \quad (\text{lens-to-wall distance})$$

$$M = b / a = 2 \quad \rightarrow \quad a = b / 2 = 30 / 2$$

$$a = 15 \text{ cm}$$

$$\text{check : } 1/f = 1/15 + 1/30 = 3/30 \quad \rightarrow \quad f = 10 \text{ cm} \quad (\text{a sensible convex lens})$$

The single word “wall” settles half the question before any algebra: a concave lens produces only virtual images, and a virtual image cannot be caught on a screen. That kills choices A and D immediately.

choice C — a convex lens with the object 15 cm away

23 Young with $d = 0.2 \text{ mm}$, $\lambda = 600 \text{ nm}$, $L = 1 \text{ m}$

$$\lambda = 6 \times 10^{-7} \text{ m}, \quad d = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}, \quad L = 1 \text{ m}$$

$$\Delta y = \lambda L / d$$

$$\Delta y = (6 \times 10^{-7} \times 1) / (2 \times 10^{-4})$$

$$\Delta y = 3 \times 10^{-3} \text{ m} = 3 \text{ mm}$$

Choice D (1 m) is there to catch a slip in the powers of ten, and choice C (30 mm) a slip of one power. The habit that protects you is to turn every length into metres in the first line, before anything is divided.

choice A — $\Delta y = 3 \text{ mm}$

24 the lens that magnifies 4 times twice over

first position – erect, magnified $4 \times$ → the image is virtual, b is negative :

$$M = |b| / a_1 = 4 \quad \rightarrow \quad |b| = 4 a_1$$

$$1/f = 1/a_1 - 1/(4 a_1) = 3 / (4 a_1) \quad \rightarrow \quad a_1 = 3 f / 4$$

second position – real, inverted, magnified $4 \times$ → b is positive :

$$M = b / a_2 = 4 \quad \rightarrow \quad b = 4 a_2$$

$$1/f = 1/a_2 + 1/(4 a_2) = 5 / (4 a_2) \quad \rightarrow \quad a_2 = 5 f / 4$$

the object was moved away by 6 cm :

$$a_2 - a_1 = 6$$

$$5f/4 - 3f/4 = 6$$

$$2f/4 = 6 \quad \rightarrow \quad f / 2 = 6$$

$$f = 12 \text{ cm}$$

$$\text{check : } a_1 = 9 \text{ cm} \quad (\text{inside } F = 12, \text{ virtual } \checkmark) \quad a_2 = 15 \text{ cm} \quad (\text{between } F \text{ and } 2F \checkmark)$$

The two positions straddle the focus, one on each side, and that is what makes the problem work: the same lens gives the same size of image twice, once virtually (inside F) and once really (between F and $2F$). The check at the end is worth writing every time — it confirms both positions land in the regions the wording described.

$f = 12 \text{ cm}$ — the object moves from 9 cm to 15 cm

25 the two lens positions for a fixed 90 cm

let the object-to-lens distance be a ; the screen is 90 cm from the object,
so the lens-to-screen distance is $b = 90 - a$

$$1/f = 1/a + 1/b$$

$$1/20 = 1/a + 1/(90 - a)$$

$$1/20 = (90 - a + a) / [a(90 - a)]$$

$$1/20 = 90 / [a(90 - a)]$$

$$a(90 - a) = 1800$$

$$a^2 - 90a + 1800 = 0$$

$$a = [90 \pm \sqrt{(8100 - 7200)}] / 2 = (90 \pm 30) / 2$$

$$a = 60 \text{ cm} \quad \text{or} \quad a = 30 \text{ cm}$$

The two roots add up to 90, and that is the whole story: the object and image distances simply swap between the two positions (30 & 60, then 60 & 30). One gives a magnified image ($M = 2$), the other a diminished one ($M = \frac{1}{2}$) — this is the standard “displacement method” for measuring a focal length.

$a = 30 \text{ cm}$ or 60 cm — two lens positions give a sharp image

26 the fringe at P when $\Delta = 1.5 \lambda$

$$\Delta = S_2P - S_1P = 1.5 \lambda$$

write it in the standard form :

$$1.5 \lambda = (1 + \frac{1}{2}) \lambda \quad \rightarrow \quad m = 1$$

this matches the destructive condition $\Delta = (m + \frac{1}{2}) \lambda$

so the two waves arrive at P exactly out of phase (half a cycle apart),
a crest of one meeting a trough of the other, and they cancel.

The rule to carry into the exam: divide the path difference by λ . A **whole** number (0, 1, 2 ...) means bright; a whole number **and a half** (0.5, 1.5, 2.5 ...) means dark. Here 1.5 is the second dark fringe out from the centre.

a dark fringe — the path difference is an odd number of half wavelengths, so the two waves arrive out of phase and interfere destructively

Part G — Review, Group C

27 red light instead of blue

$$\Delta y = \lambda L / d \quad \rightarrow \quad \Delta y \propto \lambda \quad (L \text{ and } d \text{ unchanged})$$

$$\lambda_{\text{red}} \approx 700 \text{ nm} \quad \lambda_{\text{blue}} \approx 450 \text{ nm}$$

$$\lambda_{\text{red}} > \lambda_{\text{blue}} \quad \rightarrow \quad \Delta y \text{ increases}$$

Roughly, Δy grows by the ratio $700/450 \approx 1.6$, so the pattern stretches by about half again. This is also why a white-light double slit shows a white centre with coloured edges: every colour makes its own pattern, all sharing the same central fringe.

choice A — the separation increases, because the wavelength of red light is larger

28 concave lens, $f = 20$ cm, $a = 20$ cm

for a concave lens f is negative : $f = -20$ cm

$$1/f = 1/a + 1/b$$

$$-1/20 = 1/20 + 1/b$$

$$1/b = -1/20 - 1/20 = -2/20 = -1/10$$

$$b = -10 \text{ cm} \quad (\text{negative} \rightarrow \text{virtual, on the object side})$$

$$M = |b| / a = 10 / 20 = 0.5 \quad (\text{diminished})$$

b negative with a positive $\rightarrow$ erect

Choice A is the trap: "the object is at the focal distance" sends the image to infinity only for a **convex** lens. A concave lens has no such special position — it gives the same three adjectives for every object distance, as question 4 proved.

choice B — virtual, erect and diminished, 10 cm from the lens

29 the ratio of the two fringe separations

$$\lambda_1 = 6000 \text{ Å} = 6 \times 10^{-7} \text{ m} \quad R_1 = R$$

$$\lambda_2 = 4000 \text{ Å} = 4 \times 10^{-7} \text{ m} \quad R_2 = 2R$$

the slit separation d is the same in both cases :

$$\Delta y_1 = \lambda_1 R_1 / d = 6000 R / d$$

$$\Delta y_2 = \lambda_2 R_2 / d = 4000 \times 2R / d = 8000 R / d$$

$$\Delta y_1 / \Delta y_2 = 6000 / 8000$$

$$\Delta y_1 / \Delta y_2 = 3 / 4 = 0.75$$

Because it is a ratio, the ångströms never have to be converted — the units cancel, and so does d and the unknown R . Always look for that before reaching for a calculator: two of the three letters in $\lambda R/d$ dropped out here on their own.

$$\Delta y_1 / \Delta y_2 = 3 : 4$$

30 why the image speeds up near the focus

for a convex lens $1/f = 1/a + 1/b \rightarrow b = af / (a - f)$

differentiate b with respect to a (f is a constant) :

$$db/da = [f(a - f) - af] / (a - f)^2$$

$$db/da = -f^2 / (a - f)^2$$

the object moves at a uniform speed $v = |da / dt|$

so the speed of the image is

$$v' = |db/dt| = |db/da| \times |da/dt|$$

$$v' = f^2 / (a - f)^2 \times v$$

and since $M = f / (a - f)$:

$$v' = M^2 v$$

as the object approaches F , $(a - f) \rightarrow 0$, so M grows without limit :

$\rightarrow v'$ depends on a , therefore it is VARIABLE

$\rightarrow v'$ grows as the object nears the focus, and $\rightarrow \infty$ at $a = f$

The minus sign in db/da carries its own meaning: as the object comes **towards** the lens the image runs **away** from it. And because the factor is M^2 , not M , the effect is dramatic — at $M = 10$ the image is already moving a hundred times faster than the object. This is exactly why focusing a camera on something very close is so delicate.

$v' = M^2 v$ — the image speed depends on the object position, so it is variable, and it increases without limit as the object approaches F

31 why the central fringe is always bright

the central point O lies on the perpendicular bisector of S_1S_2 , so

$$S_1O = S_2O$$

$$\Delta = S_2O - S_1O = 0 = 0 \times \lambda \quad \rightarrow \quad m = 0$$

a zero path difference means the two waves arrive exactly in phase, crest on crest, for every wavelength in the beam, so they reinforce and the centre is always the brightest fringe.

“For every wavelength” is the part worth remembering. Every other fringe sits at a position that depends on λ , so in white light they spread into colours; only the central fringe is fixed at $\Delta = 0$ for all of them, which is why it comes out white and why it is used as the reference point of the whole pattern.

because the two distances from the slits to the centre are equal, the path difference there is zero for every wavelength — the waves always arrive in phase and interfere constructively